

Lab Sheet-9

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import pandas as pd

import numpy as np

# 1. GENERATE SYNTHETIC GLOBAL SUPERSTORE SALES DATASET

np.random.seed(42)

n_samples = 300

categories = ['Technology', 'Office Supplies', 'Furniture']

regions = ['North', 'South', 'East', 'West']

data = {

    'OrderID': range(1001, 1001 + n_samples),

    'Category': np.random.choice(categories, n_samples, p=[0.4, 0.4, 0.2]),

    'Region': np.random.choice(regions, n_samples),

    'Sales': np.random.uniform(50, 1500, n_samples).round(2),

    'Discount': np.random.choice([0.0, 0.1, 0.2, 0.3, 0.5], n_samples, p=[0.3, 0.3, 0.2, 0.1, 0.1])

}

df = pd.DataFrame(data)

# Calculate Profit

def calculate_profit(row):

    base_margin = 0.25 if row['Category'] == 'Technology' else (0.15 if row['Category'] == 'Office Supplies' else 0.05)

    margin = base_margin - (row['Discount'] * 0.8)

    return round(row['Sales'] * margin, 2)

df['Profit'] = df.apply(calculate_profit, axis=1)

# Save output dataset CSV

df.to_csv('superstore_storytelling_data.csv', index=False)

# 2. KPI COMPUTATION

total_sales = df['Sales'].sum()

total_profit = df['Profit'].sum()

overall_margin = (total_profit / total_sales) * 100

avg_discount = df['Discount'].mean() * 100

print("=" * 60)

print(" EXPERIMENT 9: DATA STORYTELLING & BUSINESS INSIGHTS ")
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print("=" * 60)

print("--- KEY PERFORMANCE INDICATORS (KPIs) ---")

print(f"Total Revenue (Sales) : ${total_sales:,.2f}")

print(f"Total Net Profit      : ${total_profit:,.2f}")

print(f"Overall Profit Margin : {overall_margin:.2f}%")

print(f"Average Discount Rate : {avg_discount:.2f}%\n")

# 3. NARRATIVE BREAKDOWN BY CATEGORY

print("--- CATEGORY PERFORMANCE BREAKDOWN ---")

cat_summary = df.groupby('Category').agg(

    Total_Sales=('Sales', 'sum'),

    Total_Profit=('Profit', 'sum'),

    Avg_Discount=('Discount', 'mean')

).reset_index()

cat_summary['Profit_Margin_%'] = (cat_summary['Total_Profit'] / cat_summary['Total_Sales']) * 100

print(cat_summary.to_string(index=False))

print("\n--- BUSINESS STORYTELLING NARRATIVE ---")

print("1. Insight : Technology is driving the highest total profit margins due to lower discount dependencies.")

print("2. Challenge: High discount rates (>=30%) in Furniture/Office Supplies lead to negative profit margins.")

print("3. Action   : Cap promotional discounts at 20% to safeguard profit margins without dropping sales volume.")

print("\n[SUCCESS] Storytelling analysis complete & saved to 'superstore_storytelling_data.csv'!")

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PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL
PS C:\Users\Ashish\Desktop\Python> python exp9\ (1).py
=====
EXPERIMENT 9: DATA STORYTELLING & BUSINESS INSIGHTS
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--- KEY PERFORMANCE INDICATORS (KPIs) ---
Total Revenue (Sales) : $229,610.53
Total Net Profit      : $11,505.88
Overall Profit Margin : 5.01%
Average Discount Rate : 14.33%

--- CATEGORY PERFORMANCE BREAKDOWN ---
Category      Total_Sales  Total_Profit  Avg_Discount  ProfitMargin_%
Furniture      48651.24      -4198.35      0.166667      -8.629482
Office Supplies 83037.91       1292.46      0.158333      1.556470
Technology     97921.38      14411.77      0.119380      14.717695

--- BUSINESS STORYTELLING NARRATIVE ---
1. Insight      : Technology is driving the highest total profit margins due to lower discount dependencies.
2. Challenge     : High discount rates (>=30%) in Furniture/Office Supplies lead to negative profit margins.
3. Action       : Cap promotional discounts at 20% to safeguard profit margins without dropping sales volume.

[SUCCESS] Storytelling analysis complete & saved to 'superstore_storytelling_data.csv'!
PS C:\Users\Ashish\Desktop\Python>

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Questions & Answers

Q1. What is data storytelling? How is it different from data visualization?

Answer:

- **Data Storytelling:** The structured approach of communicating data insights using a combination of three key elements: **Data**, **Visuals**, and a narrative **Context/Story**. It explains *why* a metric changed and *what action* should be taken.
- **Data Visualization:** The visual representation of raw numerical data into graphic forms (e.g., charts, graphs).

Feature	Data Visualization	Data Storytelling
Primary Focus	Presenting facts and metrics visually	Explaining meaning, context, and cause-and-effect
Output	Dashboards, standalone charts	Coherent narrative decks, executive briefings
Goal	Enables data exploration	Drives strategic decision-making and action

Q2. Why is storytelling important in business analytics and decision-making?

Answer:

- **Bridges Technical Gaps:** Translates complex statistical outputs into plain business language for non-technical leadership.
- **Drives Engagement:** Human brains process narratives much better than standalone raw spreadsheets.
- **Accelerates Action:** Connects analytical findings directly with clear operational recommendations and ROI targets.

Q3. What are the essential components of an effective data story?

Answer:

1. **Data:** Reliable, cleaned historical metrics and quantitative evidence.
2. **Visuals:** Clear charts free of visual clutter that highlight key anomalies or trends.
3. **Narrative Context:** The baseline setting, business challenge (conflict), analytical discovery, and strategic recommendation (resolution).

Q4. How do Key Performance Indicators (KPIs) enhance business reporting?

Answer:

- **Focuses Attention:** Highlights critical performance benchmarks (e.g., *Net Profit Margin*, *Customer Acquisition Cost*) rather than overwhelming users with raw data.
- **Tracks Progress:** Measures operational performance directly against organizational strategic targets.

- **Standardizes Metrics:** Ensures alignment across cross-functional enterprise departments.
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Q5. Why should visualizations be arranged in a logical sequence while presenting analytical findings?

Answer: Arranging visualizations logically establishes a clear narrative arc (e.g., *Macro Executive Summary* $\rightarrow$ *Regional Breakdown* $\rightarrow$ *Root Cause Analysis* $\rightarrow$ *Recommendations*):

- Prevents cognitive overload for decision-makers.
 - Guides stakeholders step-by-step from problem identification to root-cause discovery.
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Q6. What factors should be considered while selecting visualizations for a business presentation?

Answer:

1. **Target Audience:** Executive leadership prefers high-level KPI cards, whereas operational managers require granular drill-down charts.
 2. **Data Relationship:**
 - *Trends over time:* Line charts.
 - *Categorical comparisons:* Bar charts.
 - *Correlations:* Scatter plots or heatmaps.
 - *Proportions:* Stacked bar charts.
 3. **Clutter Reduction:** Eliminating unnecessary gridlines, excessive colors, and 3D chart effects.
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Q7. Explain how dashboards and storytelling complement each other in Business Intelligence.

Answer:

- **Dashboards:** Provide operational monitoring platforms for tracking real-time KPIs and dynamic data filtering.
 - **Data Storytelling:** Takes snapshots of key dashboard insights, adds contextual business narrative, and formulates strategic change recommendations.
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Q8. What challenges may arise while communicating analytical insights to non-technical stakeholders?

Answer:

- **Statistical Jargon:** Overusing technical terminology (e.g., *p-values*, *eigenvalues*, *homoscedasticity*) instead of operational business metrics.
 - **Lack of Context:** Presenting percentage changes without defining baseline monetary values.
 - **Information Overload:** Presenting too many charts without a clear central thesis or call to action.
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Q9. Give two real-world examples where data storytelling has influenced business or policy decisions.

Answer:

1. **E-Commerce Logistics Optimization:** A retailer used sales data storytelling to show that free shipping thresholds increased average basket value by 35, leading management to adopt dynamic cart thresholds.
 2. **Public Health Response:** Using epidemiological data narratives during health crises to guide government resource allocations (e.g., vaccine distribution priorities).
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Q10. How can effective data storytelling improve strategic planning and organizational performance?

Answer:

- **Data-Driven Alignment:** Replaces subjective guesswork with objective quantitative insights across executive teams.
- **Risk Mitigation:** Highlights early warning indicators (e.g., margin compression due to excessive discounting) before financial losses scale.
- **Resource Optimization:** Directs operational capital toward high-return product categories and markets.